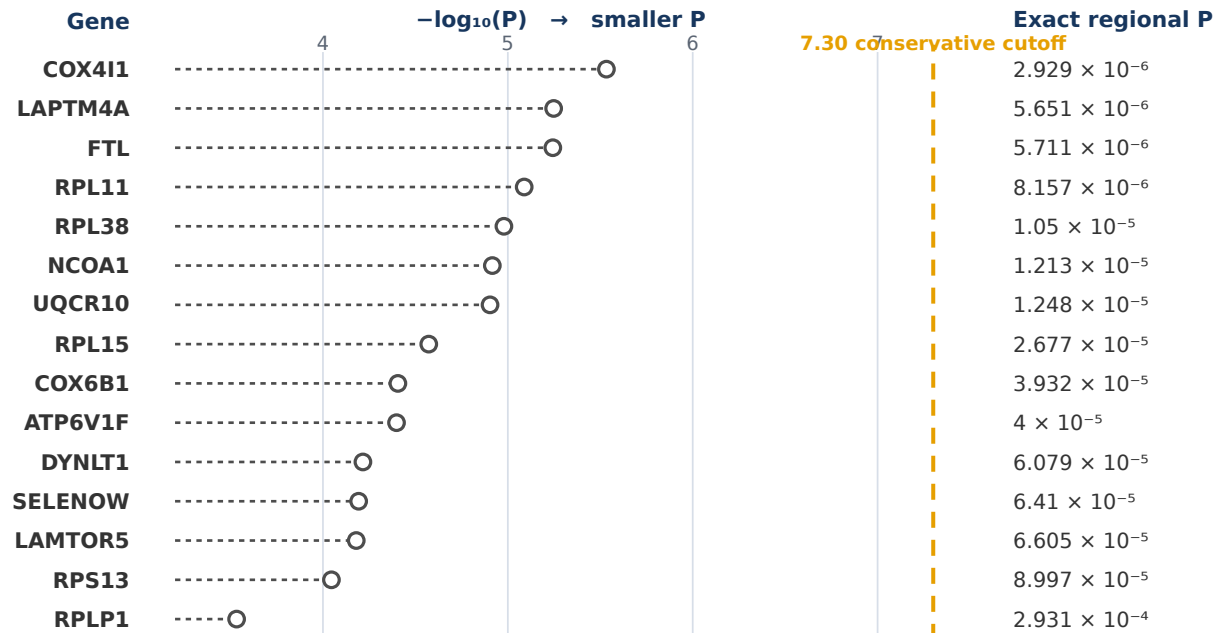


A All 19 exact values • 15 additional regions

Smallest Bellenguez AD P value within each gene's nearby region



○ 15 additional values at/above cutoff • Exact minima retained for transparent follow-up

B Four priority regions below cutoff

Gene	Exact regional P	$-\log_{10}$
◆ APOE	underflow* (stored 0)	beyond range*
◆ RPS15	4.089×10^{-30}	29.39
◆ COX7C	8.579×10^{-14}	13.07
◆ ANKRD11	1.283×10^{-11}	10.89

*APOE was too small for the file's number format; P is not literally zero.

C Why a conservative cutoff?

A genome-wide study checks millions of DNA variants.

$0.05 \div$ about 1,000,000 independent tests

$\approx 5 \times 10^{-8}$

This greatly reduces chance findings.
It is a screening rule, not a magic boundary.

D How the results guide validation

All 19 exact P values remain available for comparison.

The four below-cutoff regions are priorities for matched gene-activity and shared-variant analyses.

Nearby = 1 Mb on each side. Gene-level validation can test which gene or DNA "switch" is responsible.